Patrick Mayer – Game Developer

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SKILLS

- ***Primary Software/Languages****: C++, C,* Godot, Unity, C#, Git
- ***Secondary Software/Languages****:* WebGPU, WebGL, Win32, CMake, Java, SQL
- ***Certifications****: FDI Design Certification of Excellence*

EXPERIENCE

Florida Poly (2025 through 2026) | *Game Development Researcher*

- Designed and implemented game features such as minecart physics, a basic water shader, and boss fight scripts.
- Worked on software engineering and code refactoring throughout the project.
- Game logic was all written in C#.

PROJECTS

Spacewestern RPG (Ongoing) | *C++*

- Creating dialogue tool for connecting dialogue nodes, and for embedding a scripting language so that dialogue can have gameplay effects. Tool currently supports adding and deleting nodes, selecting and dragging nodes with selection box, and tree hierarchy refactoring for enforcing visual clarity.
- Developing 2D game engine using Win32 API, direct sound, and direct input.
- Currently supports simple audio playback and tilemaps with collisions.
- Designed talent and skill system taking inspiration from Fallout and DND.

RPG Character Creator (Ongoing) | *Godot, GDExtension, C++*

- Created tab-based UI for tweaking character attributes, skills, and traits.
- Created explanation box that gives context for a stat's in-game function when the user hovers over the stat's name.
- Stats are accurate to the game Fallout New Vegas. You cannot create a character with stats that would be invalid in the game.

Hive Mind Nebula 2 (Fall 2025) | *WebGPU*

- 6 DoF movement using quaternions with space momentum physics.
- Optimized projectile spawning and despawning using object pooling.
- Parsed and rendered 3D models in .obj and .mtl formats.

***Captain Zip (Spring* 2024)** | Godot, GDScript

- Won 1st place at the 2024 Spring FL Poly Game Jam.
- Sonic the Hedgehog inspired platformer with momentum physics and ranking system.
- Created the final level which rewards skill while still being accessible to beginners.

EXTRACURRICULAR ACTIVITIES

FL Poly Media Club

- Participated in a game jam every semester I was in school.
- Treasurer of the Media Club my sophomore year.

EDUCATION

Florida Polytechnic University, 2021-2025, Lakeland, FL
Bachelor of Science in Computer Science